Arithmetic Localization as a Necessity for Bounded Exact Dynamics

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Abstract

We study discrete-time dynamical systems defined over rational vector spaces subject to a fixed bit-length constraint. We prove that any nontrivial affine rational dynamics preserving exactness within a bounded rational state space must either collapse into eventual periodicity or implement an algebraic stabilization mechanism. We show that any such stabilization necessarily induces a canonical decomposition into prime-power components via the Chinese Remainder Theorem. Finally, we demonstrate that if the system must simultaneously maintain exact internal arithmetic and continuous external coupling, the natural completion of the state space is adelic. This establishes arithmetic localization as a structural necessity for bounded exact dynamics.

1 Introduction

Exact rational computation under recursive dynamics arises in symbolic computation, cryptography, and constrained physical models. However, rational arithmetic under iteration generically exhibits denominator growth, leading to unbounded resource consumption. This work formalizes a fundamental dichotomy: bounded exact rational dynamics are either trivial or arithmetic-localized.

2 State Space and Dynamics

Definition 2.1 (Bounded Rational State Space)**.** Let $L \in \mathbb{N}$. Define

$$S_L := \left\{ x \in \mathbb{Q}^{26} \;\middle|\; \max\left(\mathrm{bitlength}(\mathrm{num}(x_i)), \mathrm{bitlength}(\mathrm{den}(x_i))\right) \leq L \;\forall i \right\}.$$

Definition 2.2 (Affine Rational Update)**.** Let

$$f(x) = Mx + B,$$

where $M \in \mathrm{Sp}(26, \mathbb{Q})$ and $B \in \mathbb{Q}^{26}$.

Definition 2.3 (Exact Stabilized Dynamics)**.** Let $C : \mathbb{Q}^{26} \to S_L$ be an exact stabilization operator. The realized dynamics is

$$x_{n+1} = C(f(x_n)).$$

3 Denominator Growth Obstruction

Lemma 3.1 (Denominator Explosion). *If any entry of M or B has denominator divisible by $d > 1$, then for generic $x_0 \in \mathbb{Q}^{26}$, the bitlength of at least one coordinate of $f^n(x_0)$ grows at least*

$$\Omega(n \log_2 d).$$

Proof. Iterated rational multiplication accumulates denominators multiplicatively unless exact cancellation occurs, which is nongeneric. □

4 The Finite-State Trap

Theorem 4.1 (Finite-State Trap for Bounded Exact Dynamics). *Let $A \subseteq S_L$ be a nonempty invariant set.*

1. *Without stabilization ($C = \mathrm{Id}$), all orbits in A are eventually periodic.*
2. *Sustaining non-periodic dynamics requires C to implement algebraic reduction.*

Proof. Since S_L is finite, any deterministic map without reduction induces a finite-state automaton, hence eventual periodicity. □

5 Arithmetic Localization

Corollary 5.1 (Prime-Power Inevitability). *If stabilization is achieved via exact modular or residue-number reduction, then the state representation decomposes canonically into independent prime-power channels.*

Proof. By the Chinese Remainder Theorem,

$$\mathbb{Z}/M\mathbb{Z} \cong \prod_{p^k \| M} \mathbb{Z}/p^k\mathbb{Z}.$$

□

6 Adelic Completion

Axiom 6.1 (Place Completeness). The system must maintain exact internal arithmetic stability and continuous external coupling.

Proposition 6.1 (Adelic Packaging). *Under Place Completeness, the natural state space completion is*

$$\mathbb{A}_{\mathbb{Q}}^{26} = \mathbb{R}^{26} \times \prod_p{}' \mathbb{Q}_p^{26},$$

with compatibility enforced by the product formula

$$\prod_v |x|_v = 1.$$

7 Floquet Dynamics

7.1 Exact Rational Floquet Operator

Let

$$U_F = e^{-iH_2T_2}e^{-iH_1T_1}, \quad H_1, H_2 \in M_n(\mathbb{Q}), \quad T_1, T_2 \in \mathbb{Q}.$$

Exact iteration of U_F generically produces denominator growth unless stabilized.

7.2 Prime-Power Floquet Decomposition

If stabilization is implemented modulo M, then

$$U_F \bmod M \cong \prod_{p^k \| M} U_F^{(p^k)}.$$

8 Floquet Winding Number

Let $U(k)$ be a stabilized Floquet operator depending smoothly on crystal momentum $k \in S^1$. Define

$$W = \frac{1}{2\pi i}\int_{S^1} \mathrm{Tr}\left(U^{-1}(k)\,\partial_k U(k)\right)\,dk.$$

Proposition 8.1. *Under arithmetic stabilization, $W \in \mathbb{Z}$ and is invariant under exact iteration.*

9 Noncommutative Geometry

9.1 Spectral Triple

The stabilized dynamics defines a spectral triple

$$(\mathcal{A}, \mathcal{H}, D), \quad D = -i\partial_k,$$

where $\mathcal{A}$ is the algebra generated by stabilized Floquet operators.

9.2 K-Theory Pairing

Let P be the spectral projection onto a Floquet band. The index pairing is

$$\mathrm{Index}(PUP) = \frac{1}{2\pi i}\int_{S^1} \mathrm{Tr}\left(P\,U^{-1}(k)\,\partial_k U(k)\right)\,dk.$$

10 Numerical Appendix

Listing 1: Exact rational iteration demonstrating denominator growth

```
from fractions import Fraction

x = Fraction(1,1)
```

```python
for n in range(10):
    x = Fraction(3,2) * x
    print(n, x.denominator.bit_length())
```

11 Conclusion

Bounded exact rational dynamics face a strict dichotomy: periodic collapse or arithmetic localization. Prime-power decomposition is structurally necessary, and adelic architecture provides the unique coherent global-local completion.